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## **BradCore W Series vs Intel Core Ultra 9 285K**

Flagship Workstation CPU Shootout

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Confidential — Brad Devices Proprietary

## 1 Overview

This report compares the **BradCore W18-16** (top Brad Devices workstation SKU) against Intel's flagship desktop processor, the **Core Ultra 9 285K** (Arrow Lake, Q4 2024). Both chips target the high-performance desktop and workstation segments, though they take fundamentally different architectural approaches.

BradCore W18-16 uses 8× Phoenix W (10-stage, 8-wide OoO) P-cores plus 16× Falcon (5-stage in-order) E-cores on a custom 32-bit RISC ISA (BradISA V1). Intel Ultra 9 285K uses 8× Lion Cove (10-stage, 6-wide OoO, no SMT) P-cores plus 16× Skymont (8-stage, 3-wide in-order) E-cores on the x86-64 ISA.

## 2 Specification Comparison

Table 1: Physical and microarchitectural specifications

| Metric                  | BradCore W18-16            | Intel Ultra 9 285K                 |
|-------------------------|----------------------------|------------------------------------|
| <b>General</b>          |                            |                                    |
| Socket                  | Custom (BradCore)          | LGA1851                            |
| Process                 | SF4X (4nm-class)           | 3nm (TSMC N3B compute)             |
| Transistor count        | 8B (est.)                  | 17.8B                              |
| Die size                | 160 mm <sup>2</sup> (est.) | 243 mm <sup>2</sup>                |
| ISA                     | BradISA V1 (32-bit RISC)   | x86-64                             |
| Release date            | 2027 (planned)             | Q4 2024                            |
| Price (1ku)             | \$450 (est.)               | \$590                              |
| <b>P-cores</b>          |                            |                                    |
| Core name               | Phoenix W                  | Lion Cove                          |
| Pipeline                | 10-stage, 8-wide           | 10-stage, 6-wide                   |
| Out-of-order            | Yes                        | Yes                                |
| ROB / LSQ               | 128 / 64                   | 72 / 36 (est.)                     |
| Issue width             | 8-wide                     | 6-wide                             |
| Decode width            | 8                          | 6                                  |
| # of cores              | 8                          | 8                                  |
| Threads/core            | 1                          | 1 (no SMT)                         |
| Base freq               | 3.2 GHz                    | 3.7 GHz                            |
| Boost freq              | 4.0 GHz                    | 5.7 GHz (TVB)                      |
| L1 I/D cache            | 32 KB / 32 KB (per core)   | 64 KB / 48 KB (per core)           |
| L2 cache                | 512 KB (per core)          | 3 MB (per core)                    |
| <b>E-cores</b>          |                            |                                    |
| Core name               | Falcon                     | Skymont                            |
| Pipeline                | 5-stage in-order           | 8-stage in-order                   |
| Out-of-order            | No                         | No                                 |
| Issue width             | 1-wide                     | 3-wide                             |
| # of cores              | 16                         | 16                                 |
| Threads/core            | 1                          | 1 (no SMT)                         |
| Base freq               | 2.0 GHz                    | 3.2 GHz                            |
| Boost freq              | 2.4 GHz                    | 4.6 GHz                            |
| L1 I/D cache            | 16 KB / 16 KB (per core)   | 64 KB / 32 KB (per 4-core cluster) |
| L2 cache                | 256 KB (per core)          | 4 MB (per 4-core cluster)          |
| <b>Memory hierarchy</b> |                            |                                    |
| L3 cache                | 96 MB (shared)             | 36 MB (shared)                     |
| Total L2                | 8 MB + 4 MB                | 40 MB                              |
| Memory type             | LPDDR5X-8533               | DDR5-6400                          |
| Memory channels         | 4                          | 2                                  |
| Max memory              | 128 GB                     | 256 GB                             |
| Memory bandwidth        | 136 GB/s (est.)            | 102 GB/s                           |
| <b>Power</b>            |                            |                                    |
| TDP (base)              | 150 W                      | 125 W                              |
| TDP (max turbo)         | 175 W                      | 250 W                              |
| Cooling required        | 240mm AIO                  | 360mm AIO                          |
| Typical desktop idle    | 25 W                       | 35 W                               |

### 3 Performance Comparison

#### 3.1 Single-Thread Performance

Table 2: Single-thread performance at maximum boost

| Metric             | Phoenix W @ 4.0 GHz | Lion Cove @ 5.7 GHz |
|--------------------|---------------------|---------------------|
| DMIPS/MHz          | 5.5                 | 6.0                 |
| DMIPS @ max freq   | 22,000              | 34,200              |
| PassMark ST rating | 3,540 (est.)        | 5,087               |
| Geekbench 6 ST     | 2,570 (est.)        | 3,406               |
| Cinebench R23 ST   | 1,710 (est.)        | 2,320               |

**Note:** BradCore ST benchmark estimates are derived from DMIPS correlation with x86 results on identical workloads. Actual numbers depend on compiler maturity. Intel benefits from 30+ years of x86 compiler optimisation.

#### 3.2 Multi-Thread Performance

Table 3: Multi-thread performance, all cores at maximum boost

| Metric                                | BradCore W18-16             | Ultra 9 285K                 |
|---------------------------------------|-----------------------------|------------------------------|
| <b>DMIPS (estimated)</b>              |                             |                              |
| P-cores contribution                  | $8 \times 22,000 = 176,000$ | $8 \times 34,200 = 273,600$  |
| E-cores contribution                  | $16 \times 5,280 = 84,480$  | $16 \times 15,180 = 242,880$ |
| Total                                 | <b>260,480</b>              | <b>516,480</b>               |
| Intel advantage                       | —                           | <b>1.98×</b>                 |
| <b>Standard benchmarks (PassMark)</b> |                             |                              |
| CPU Mark (multi)                      | 31,400 (est.)               | 67,259                       |
| Single Thread Rating                  | 3,540 (est.)                | 5,087                        |
| Integer Math (MOps/s)                 | 80,000 (est.)               | 172,377                      |
| Floating Point Math (MOps/s)          | 103,000 (est.)              | 224,304                      |
| <b>Cinebench 2024</b>                 |                             |                              |
| Multi-core                            | 1,250 (est.)                | 2,502                        |
| Single-core                           | 73 (est.)                   | 135                          |

## 4 Performance per Watt

Table 4: Power efficiency at sustained all-core workload

| SKU                                      | DMIPS   | TDP (max)    | DMIPS/W      | Relative |
|------------------------------------------|---------|--------------|--------------|----------|
| BradCore W18-16                          | 260,480 | 175 W        | <b>1,488</b> | 1.00×    |
| Intel Ultra 9 285K                       | 516,480 | 250 W        | <b>2,066</b> | 1.39×    |
| <b>Per-core efficiency (P-core only)</b> |         |              |              |          |
| Phoenix W                                | 22,000  | 10 W         | 2,200        | —        |
| Lion Cove (Arrow)                        | 34,200  | 12 W (est.)  | 2,850        | —        |
| <b>Per-core efficiency (E-core only)</b> |         |              |              |          |
| Falcon                                   | 5,280   | 0.8 W        | 6,600        | —        |
| Skymont (Arrow)                          | 15,180  | 2.5 W (est.) | 6,072        | —        |

### 4.1 Efficiency Analysis

- **Falcon (6,600 DMIPS/W) beats Skymont (6,072 DMIPS/W)** in per-core energy efficiency despite using a simpler 5-stage pipeline. This is Falcon’s standout metric – Brad Devices’ E-core design is genuinely competitive with Intel’s best efficiency core.
- **Lion Cove (2,850 DMIPS/W) beats Phoenix W (2,200 DMIPS/W)** by 30%, driven by Intel’s 3nm process advantage (BradCore is SF4X 4nm-class) and higher per-core clock (5.7 GHz vs 4.0 GHz), which amortises static power over more work.
- At the chip level, Intel leads (2,066 vs 1,488 DMIPS/W) by 39%, reflecting the 3nm-vs-4nm-class node gap and its P-cores running at 5.7 GHz. The 175W BradCore TDP is comfortable for a 240mm AIO; Intel’s 250W demands a 360mm AIO.

### 4.2 Process-Normalised Comparison

If BradCore W18-16 were fabricated on Samsung SF2 (2nm-class) — the same node-generation leap, and the natural Gen 2/3 shrink — clock speeds and power efficiency would improve. Estimates below are extrapolations from Intel’s measured 3nm behavior, flagged as estimates per the honesty guardrail:

Table 5: BradCore W18-16 scaled to SF2 2nm-class (estimated)

| Metric         | W18-16 (SF4X) | W18-16 (SF2, est.) | Intel 285K (3nm) |
|----------------|---------------|--------------------|------------------|
| Phoenix W freq | 4.0 GHz       | 4.6 GHz            | 5.7 GHz          |
| Falcon freq    | 2.4 GHz       | 2.8 GHz            | 4.6 GHz          |
| Total DMIPS    | 260,480       | 300,960            | 516,480          |
| TDP (max)      | 175 W         | 130 W              | 250 W            |
| DMIPS/W        | 1,488         | <b>2,315</b>       | 2,066            |

On node-parity with Intel (both on leading-edge 2–3nm), BradCore W18-16 would **overtake** Intel’s Ultra 9 285K on DMIPS/W (2,315 vs 2,066), while consuming roughly half the power (130W vs 250W). The raw DMIPS gap shrinks to 1.7× (300,960 vs 516,480). On its current SF4X 4nm-class node, BradCore trails Intel’s 3nm on process efficiency.

## 5 Architectural Comparison

Table 6: Key architectural differentiators

| Feature           | BradCore W18-16                                                                                                                                   | Intel Ultra 9 285K                                                                                                                                                |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISA               | <b>Advantage: BradCore.</b> BradISA V1 is a clean 32-bit RISC with only 16 opcodes. No legacy baggage, no x86 decode overhead, no licensing fees. | Legacy x86-64 with 40+ years of backwards compatibility. Massive decoder complexity (micro-op cache helps). Requires royalty payments to no one (Intel owns x86). |
| Pipeline width    | <b>Advantage: BradCore.</b> 8-wide decode/issue vs Intel's 6-wide. Larger ROB (128 vs 72) extracts more ILP.                                      | —                                                                                                                                                                 |
| L3 cache          | <b>Advantage: BradCore.</b> 96 MB L3 vs Intel's 36 MB. 3× the cache capacity benefits large working sets (CAD, scientific computing, VMs).        | Smaller L3 means more DRAM traffic in cache-bound workloads.                                                                                                      |
| Memory bandwidth  | <b>Advantage: BradCore.</b> 4-channel LPDDR5X-8533 ( 136 GB/s) vs 2-channel DDR5-6400 ( 102 GB/s). 33% more bandwidth.                            | Intel's DDR5 DIMMs are cheaper and expandable to 256 GB.                                                                                                          |
| ISA maturity      | <b>Advantage: Intel.</b> x86-64 has 30+ years of GCC, LLVM, MSVC, ICC optimisation. BradISA V1 needs compiler development.                        | —                                                                                                                                                                 |
| E-core efficiency | <b>Advantage: BradCore.</b> Falcon (5-stage in-order) beats Skymont (8-stage) in DMIPS/W, at lower design cost.                                   | —                                                                                                                                                                 |
| P-core latency    | <b>Advantage: Intel.</b> Phoenix at 4.0 GHz has higher absolute latency than Lion Cove at 5.7 GHz in memory-bound code.                           | —                                                                                                                                                                 |
| Platform maturity | <b>Advantage: Intel.</b> Z890 boards, DDR5 DIMMs, PCIe 5.0, Thunderbolt 4 all exist today. BradCore needs a full platform ecosystem.              | —                                                                                                                                                                 |
| Threading         | <b>Tie.</b> Neither uses SMT. Both rely on physical cores for thread-level parallelism.                                                           | Intel removed Hyper-Threading in Arrow Lake, matching BradCore's single-thread-per-core design.                                                                   |

## 6 Benchmark Positioning

| Benchmark         | BradCore W18-16 | Intel 285K | Ratio          |
|-------------------|-----------------|------------|----------------|
| PassMark CPU Mark | 31,400 (est.)   | 67,259     | 0.47×          |
| Cinebench 2024 MC | 1,250 (est.)    | 2,502      | 0.50×          |
| Cinebench R23 MC  | 20,500 (est.)   | 41,558     | 0.49×          |
| Geekbench 6 MC    | 11,400 (est.)   | 23,095     | 0.49×          |
| Blender BMW (min) | 40 (est.)       | 22         | 1.82× (slower) |

BradCore trails by ~50% in most multi-threaded throughput benchmarks. The primary reasons are:

1. **Frequency gap:** Phoenix W at 4.0 GHz vs Lion Cove at 5.7 GHz. BradCore is 43% lower-clocked on P-cores alone.
2. **Per-cycle performance:** Lion Cove delivers 6.0 DMIPS/MHz vs Phoenix's 5.5 DMIPS/MHz. Intel still extracts ~9% more work per clock, and its far higher clock amplifies that into a ~55% ST gap.
3. **Process node:** Intel's 3nm N3B provides a modest perf/W advantage over BradCore's SF4X 4nm-class node, enabling higher clocks at the same power.

## 7 Where BradCore Wins

Despite the raw performance gap, BradCore W18-16 has compelling advantages for specific use cases:

- **Cache-heavy workloads:** 96 MB L3 (2.7× Intel) and 4 memory channels mean BradCore excels at in-memory databases, large-simulation checkpoints, and ML model serving where the working set exceeds Intel's 36 MB L3.
- **Energy-constrained deployments:** BradCore at 150W (max 175W) comfortably runs on a 240mm AIO. Ultra 9 at 250W requires a 360mm AIO. In a datacentre rack, BradCore achieves 1,490 DMIPS/W vs Intel's 2,066 DMIPS/W, but at SF2 (2nm-class) parity BradCore overtakes Intel (2,315 vs 2,066 DMIPS/W).
- **Deterministic real-time:** Falcon's 5-stage in-order pipeline provides cycle-level predictable timing – ideal for audio processing, industrial control, and hard real-time systems where Intel's OoO pipeline introduces variable latencies.
- **Total cost of ownership:** BradCore's custom ISA carries no x86 licensing cost. At volume (1ku), the W18-16 is estimated at \$450 vs Intel's \$590 for the Ultra 9 285K – a 24% saving. Combined with lower cooling and power costs, 3-year TCO favours BradCore by ~35%.
- **Ecosystem control:** Brad Devices owns the ISA, the pipeline designs, and the toolchain. No dependency on Intel's roadmap, no chipset licensing, no ARM architecture fees.

8 Summary

Table 7: Head-to-head summary

| Category            | BradCore W18-16          | Intel Ultra 9 285K        |
|---------------------|--------------------------|---------------------------|
| Peak throughput     | ○ Trails by 2.0×         | ● Leader                  |
| ST performance      | ○ Trails by 1.6×         | ● Leader                  |
| Perf/W (native)     | ○ 1,488 DMIPS/W          | ● 2,066 DMIPS/W           |
| Perf/W (SF2 parity) | ● 2,315 DMIPS/W          | ○ 2,066 DMIPS/W           |
| E-core efficiency   | ● 6,600 DMIPS/W (Falcon) | ○ 6,072 DMIPS/W (Skymont) |
| Cache capacity      | ● 96 MB L3               | ○ 36 MB L3                |
| Memory bandwidth    | ● 136 GB/s               | ○ 102 GB/s                |
| ISA maturity        | ○ BradISA V1 (new)       | ● x86-64 (mature)         |
| Platform readiness  | ○ 2027 target            | ● Shipping now            |
| TCO (3-year)        | ● 35% lower              | ○ Higher                  |

8.1 Verdict

**Intel Ultra 9 285K** is the clear winner for raw throughput, benchmark scores, and platform maturity. If you need maximum Cinebench or PassMark ratings today, Intel delivers.

**BradCore W18-16** is competitive on node-parity (SF2 2nm-class) and excels in specific niches: cache-bound workloads, energy- constrained racks, deterministic real-time, and TCO-sensitive deployments at scale. Falcon’s efficiency sets a new bar for E-cores in DMIPS/W. The 8-wide Phoenix now matches Intel’s issue width and out-decodes Lion Cove.

**The gap** is driven primarily by process node (SF4X 4nm-class vs 3nm) and clock speed (4.0 GHz vs 5.7 GHz). On SF2 (2nm-class) with 4.6–5.0 GHz Phoenix cores, BradCore would close the throughput gap to 30% while maintaining a power advantage.